Exp. No: 4

Analysis of Continuous Charge Distribution in Electric Field (Line, Surface and Volume Charge Density)

Aim:

To write program for continuous Charge Distribution in electric field. The different types of charge density are (a) Linear Charge Density, (b) Surface Charge Density, (c) Volume Charge Density using MATLAB.

Objectives:

1. To compute line, surface and volume charge densities for a given continuous charge distribution.

2. To analyze how electric field intensity varies for each type of charge distribution.

3. To compare the computed charge densities against theoretical or analytical expressions.

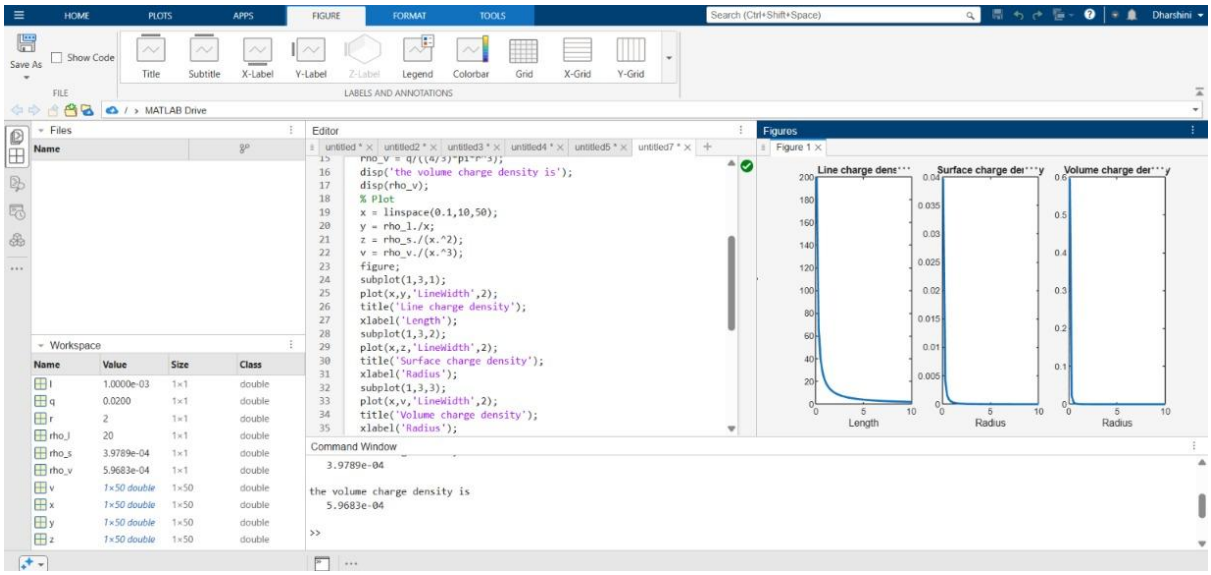
Program:

Test case (i) Determination of line, surface and volume charge density for given charge, length and radius

```
q=input("Enter the charge value
is:"); l=input("Enter the length
value is:");
% Line
charge
density
rho_l = q/l;
disp('the line charge density
is'); disp(rho_l);
% Surface charge
density r=input("Enter
the radius value is:");
rho_s = q/(4*pi*r^2);
disp('the surface charge
density is'); disp(rho_s);
% Volume
charge
density rho_v
=
q/((4/3)*pi*r^
```

```
3);
disp('the volume charge
density is'); disp(rho_v);
% Plot
x = linspace(0.1,10,50);
y = rho_l./x;
z = rho_s./(x.^2);
v =
rho_
v./(x.
^3);
figur
e;
subpl
ot(1,
3,1);
plot(x,y,'LineWid
th',2); title('Line
charge density');
xlabel('Length');
subplot(1,3,2);
plot(x,z,'LineWidt
h',2);
title('Surface
charge density');
xlabel('Radius');
subplot(1,3,3);
plot(x,v,'LineWid
th',2);
title('Volume
charge density');
xlabel('Radius');
```

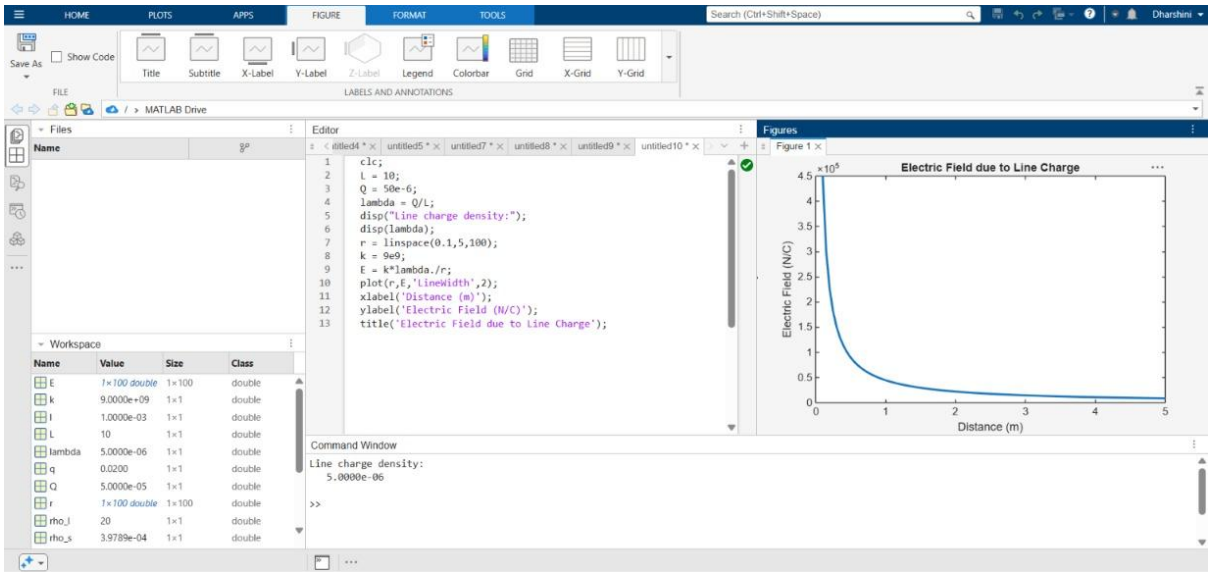
OUTPUT



Test case (ii) A long wire of length L = 10 m carries a uniform charge of Q = 50 µC. Calculate the line charge density and plot electric field variation

```
L = 10;
Q = 50e-6;
lambda = Q/L;
disp("Line charge
density:");
disp(lambda);
r =
linspace(0.
1,5,100); k
= 9e9;
E = k*lambda./r;
plot(r,E,'LineWidth'
,2);
xlabel('Distance
(m)');
ylabel('Electric
Field (N/C)');
title('Electric Field due to Line Charge');
```

OUTPUT

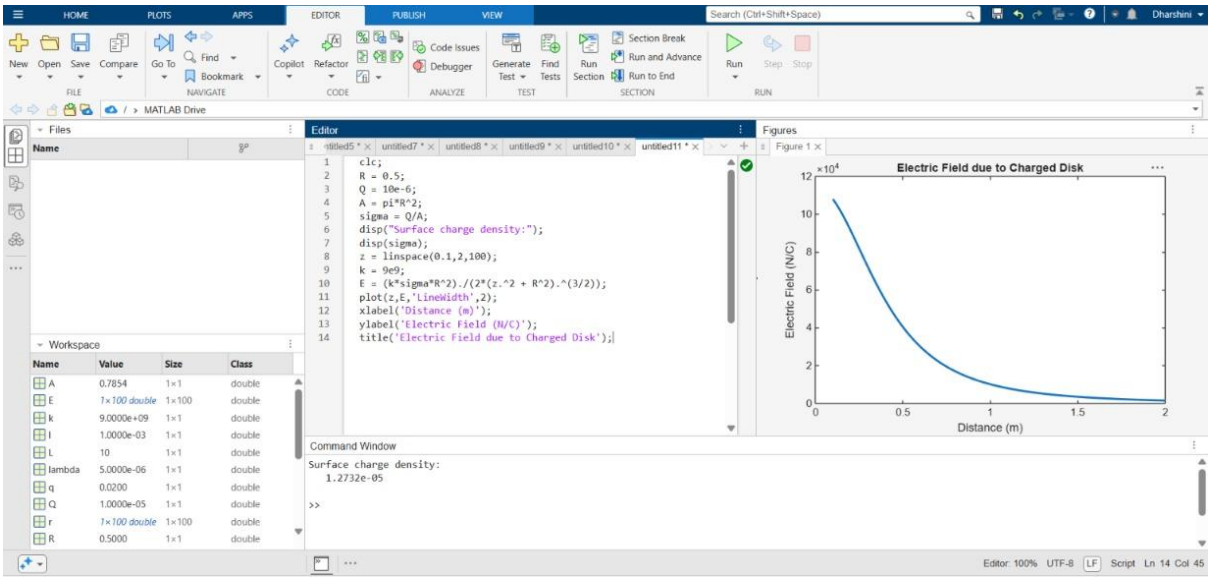


Test case (iii) A uniformly charged disk of radius R = 0.5 m with total charge Q = 10 μC.

Calculate surface charge density and electric field

```
R = 0.5;
Q = 10e-6;
A =
pi*R
^2;
sigm
a =
Q/A;
disp("Surface
charge density:");
disp(sigma);
z =
linspace(0.
1,2,100); k
= 9e9;
E = (k*sigma*R^2)./(2*(z.^2 +
R^2).^(3/2));
plot(z,E,'LineWidth',2);
xlabel('Distance
(m)');
ylabel('Electric
Field (N/C)');
title('Electric Field due to Charged Disk');
```

OUTPUT

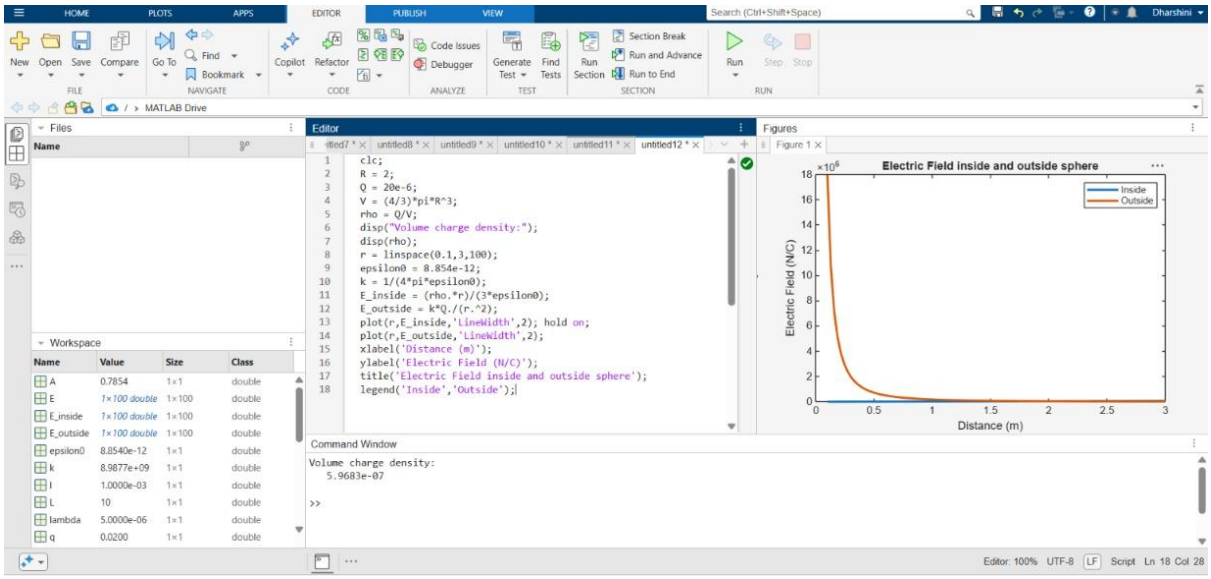


Test case (iv) A uniformly charged sphere of radius R = 2 m with total charge Q = 20 μC. Calculate volume charge density and electric field inside and outside 3

```
R = 2;
Q = 20e-6;
V =
(4/3)*pi
*R^3;
rho =
Q/V;
disp("Volume charge
density:"); disp(rho);
r =
linspace(0.
1,3,100);
epsilon0 =
8.854e-12;
k =
1/(4*pi*ep
silon0);
E_inside =
(rho.*r)/(3*epsilon0);
E_outside = k*Q./(r.^2);
plot(r,E_inside,'LineWidth',
2); hold on;
plot(r,E_outside,'LineWidth',
2); xlabel('Distance (m)');
```

```
ylabel('Electric Field (N/C)');
title('Electric Field inside and
outside sphere');
legend('Inside','Outside');
```

OUTPUT



S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Identify physical quantities and force models	Identify physical quantities and force models	Correctly identifies charge/current, separation, medium and force direction for both cases.	Identifie s most quantities and models correctl y.	Needs guidance in selecting the correct relation or direction.	Uses incorrect force models or quantities.	25	2.5	
2	Calculate electric and magnetic forces	Calculate electric and magnetic forces	Implements Coulomb-force and parallel-current force calculations correctly with units.	Calculat ions are mostly correct with minor errors.	Some calculations are correct but require assistance.	Force calculations are invalid or absent.	25	2.5	

3	Evaluate force versus separation	Evaluate force versus separation	Correctly varies separation, plots trends and checks theoretical distance dependence.	Trend and plots are correct with minor omissio ns.	Limited sweep or incomplete trend interpretatio n.	Does not establish a valid relationship.	30	3	
4	Evaluate force versus separation	Evaluate force versus separation	Correctly varies separation, plots trends and checks theoretical distance dependence.	Trend and plots are correct with minor omissio ns.	Limited sweep or incomplete trend interpretation.	Does not establish a valid relationship.	20	2	
Tota l							100	10	